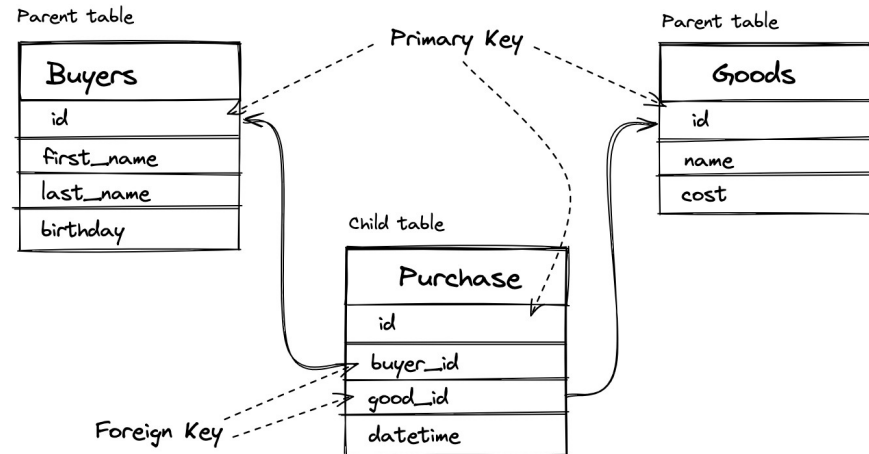


# Lecture 11

## Relational Databases



# Announcements

- ❖ Great job on Lab 3!
- ❖ Course project details coming soon
- ❖ Grades for labs will be up this week

# Last Time

- ❖ Modes and bimodal distributions
- ❖ Aggregations using split-apply-combine framework
- ❖ Interpreting statistics broken out by subgroups

# Today

- ❖ Talk about databases

# Motivation

Let's imagine a student grades dataset with the following variables:

- ❖ Student ID, Current Major, Student Phone #, Term, Course, Professor, Room, Grade

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- ❖ Each row represents a course a student took during a particular term

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Brainstorm:

- ❖ What are some the downsides of storing data this way?

# Potential Downsides

- ❖ Redundancy

- Each row for a student ID will contain Current Major and Student Phone #

- ❖ Data Management

- If a student changes their major, do we want to have to update all of their records?
  - If a first semester student drops all of their classes, do they disappear from our data?

- ❖ Privacy

- Should checking a student's major also give access to their Phone #?
  - Should checking which courses a student has taken always give access to grades?



# Relational Database

Idea: store data in multiple connected data frames (tables)

- ❖ The unit of observation will often differ between the tables
- ❖ Combine tables together only when we need info from both

# Decomposing Data

| Student ID | Current Major | Student Phone # |
|------------|---------------|-----------------|
| 001        | CS            | 555-6789        |
| 002        | DS            | 555-1234        |
| ⋮          | ⋮             | ⋮               |

**Student Info Table**

| Student ID | Term | Course | Professor     | Room      | Grade |
|------------|------|--------|---------------|-----------|-------|
| 001        | S26  | CS101  | Bonham-Carter | Alden 109 | A-    |
| 001        | S26  | CS105  | Mullins       | Alden 101 | B+    |
| ⋮          | ⋮    | ⋮      | ⋮             | ⋮         | ⋮     |

**Course Records Table**

# Decomposing Data

**Primary keys:** column names that uniquely identify a record in a table

- ❖ Determine the unit of observation

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# Decomposing Data

What's the relationship between these two tables?



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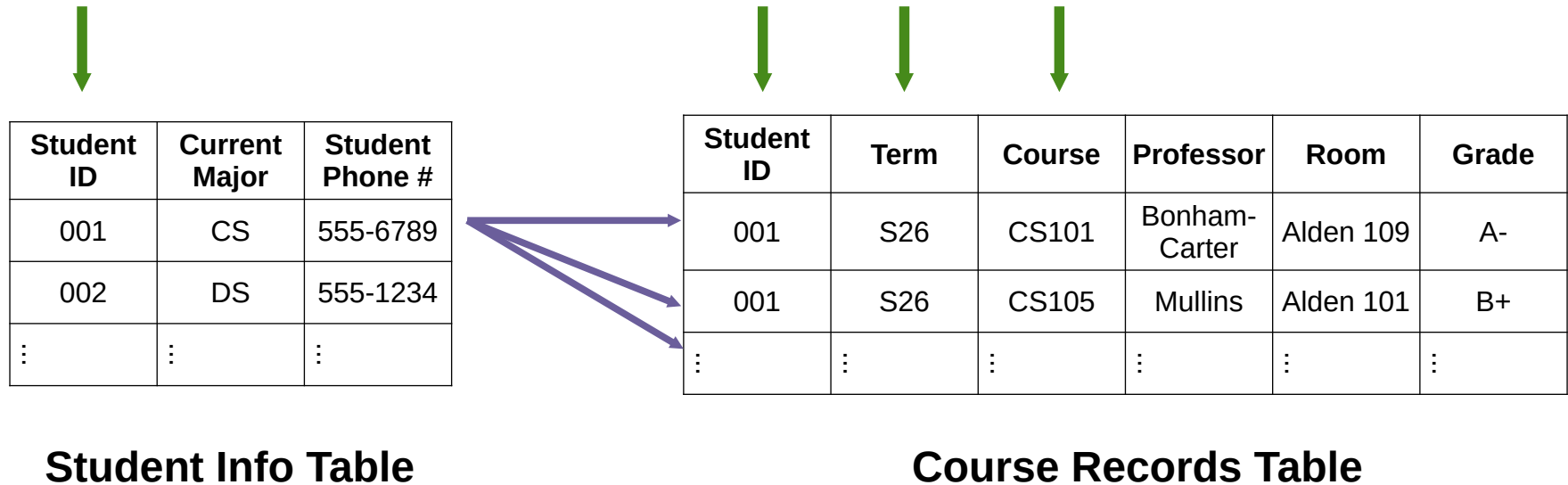
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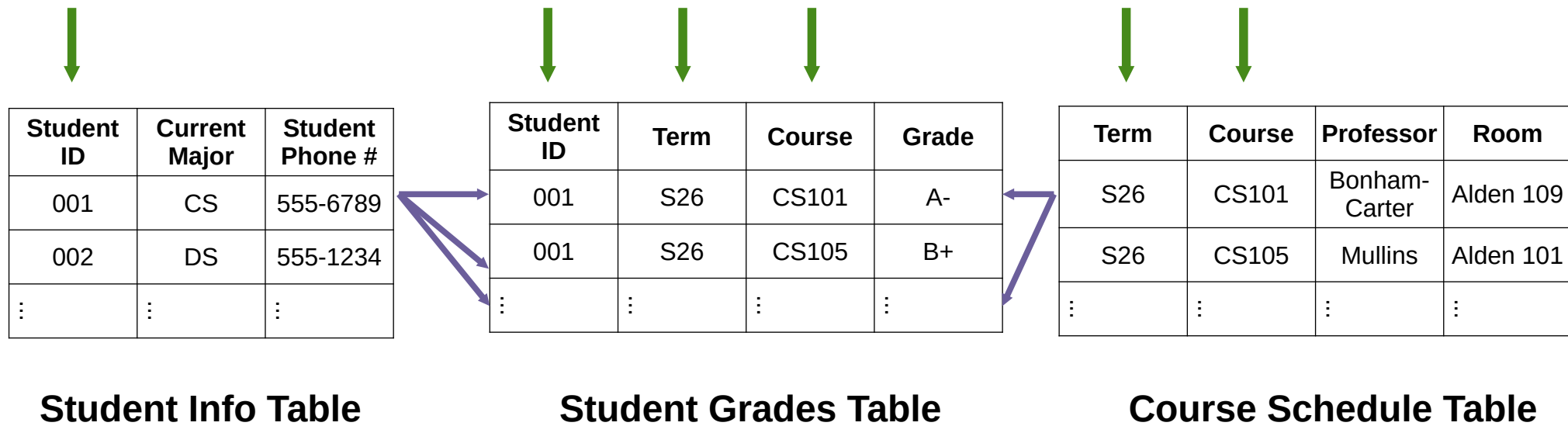
# Decomposing Data

What's the relationship between these two tables?

❖ One-to-Many



# Decomposing Data Even More



# Next Time

- ❖ Begin thinking about data wrangling
- ❖ You won't always have pristine datasets to work with!